

Gate E0 report — run_v1.22_r1 (Paper IV v1.22-r1)

Verdict: **PASS**. Evidence: 20_EVIDENCE/E0/. Auditor: Claude Opus 5.5, same session as run_v1.2_r1/run_v1.21_r1; no Lean executed.

E0 — Identity, integrity and preservation (initial and final)

Script: e0_v122.py (auditor-written, read-only). Runs: E0_INITIAL.json (2026-10-01T09:54Z, before any review) and E0_FINAL.json (2026-10-01T10:35Z, after E2–E8). **The two checks objects are identical.**

Check	Result
AUDIT_TARGET_v1.22.json sha256 cbe0fd6e... matches its sidecar	True
Candidate manifest MANUSCRIPT_REVIEW_MANIFEST.json (d9e559a3...) sha and bytes as bound	True / True
Package PAPER_IV_v1.22_REVIEW_PACKAGE_r1.zip (2b87b746...) sha and bytes as bound; sidecar	True / True / True
Mandate (7dee0923...) and frozen source manifest (fb459343...) as bound	True
Earlier evidence: run_v1.2_r1 ZIP, run_v1.21_r1 ZIP, Lean source ZIP (cb274145...), annex ZIP (2847a422...)	all sha and bytes True
Manifest vs directory: 223 files, 0 mismatches	OK
ZIP: 225 members (= 223 + manifest + sidecar), testzip OK, no unsafe paths, no hash mismatch, no extra or missing member	OK
Other ZIPs in the candidate directory	none (single package; E8-03)
Lean cut lean_piv-v12-fb459343d234: 615/615 manifest entries match, no extra files	OK
run_v1.2_r1: 1097 manifest files unchanged on disk; ZIP sidecar matches; testzip OK	OK
run_v1.21_r1: 340 manifest files unchanged on disk; ZIP sidecar matches; testzip OK	OK

The PDFs have 72 pages (EN) and 73 pages (ES).

E0 verdict: PASS. The identity is complete and matching, so the review was allowed to start. Neither the inputs nor the historical evidence changed during the review.

PDF generated by pandoc (HTML) + PyMuPDF Story from the Markdown source; not compiled from TeX. The TeX file alongside is pandoc's standalone conversion of the same Markdown.